# INTRODUCTION TO PROBLEM SOLVING AND ALGORITHM

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# LEARNING OUTCOMES

- Learners will be able to clearly identify and define a problem in logical terms
- Learners will understand what an algorithm is and its role in breaking down problems into solvable steps.
- They will be able to represent an algorithm visually using flowcharts.

## BASIC COMPUTER OPERATION

A Computer can:

1. Input: receive information.
2. Output: produce information.
3. Process:
   - perform arithmetic.
   - assign value to piece of data.
   - compare two piece of information and select one.
   - repeat a group of actions.

# GENERAL PROBLEM SOLVING CONCEPTS

1. Problem Identification
2. Problem Breakdown
3. Plan the solution
4. Implement the code

# DIFFICULTIES IN PROBLEM SOLVING

- Lack of Clarity
- Overwhelming Complexity
- Limited Resources
- Trail and Error

## STEPS OF DEVELOPING A PROGRAM

- Define the problem.
- Outline the solution.
- Develop the outline into an algorithm
- Code the algorithm into a sepcific programming language.
- Run the program on the computer.

## PROBLEM DEFINING DIAGRAM

| Input | Processing | Output |
| --- | --- | --- |
| | | |

## PROBLEMS

Create probelm defining diagram for:

- A coffee machine takes coffee ingredients and delivers coffee.
- A program calculate the area of rectangle.
- A program determines if a number is odd or even.
- An ATM allows user to withdraw cash.
- A self check-out system in supermarket scans item and calculates the total item.

# ALGORITHM

"An algorithm is a step-by-step set of instructions to solve a specific problem."

**Importance of Algorithm in Programming:**

- Algorithms break down complex problems into simple, logical steps, making them easier to implement in code.
- Algorithms can be adapted and reused across different programs and applications.

# EXAMPLE OF ALGORITHM

**Problem:** Finding largest of two numbers.
**Algorithm:**

- Start.
- Input two numbers, num1 and num2.
- If num1 > num2, output num1 is larger.
- Else, output num2 is larger.
- End.

# Algorithms in Real Life

Meta

NETFLIX

tinder

## Chess Engines and Their Algorithms

Stockfish (Minimax with Alpha-Beta Pruning)

Leela Chess Zero (Monte Carlo Tree Search + Deep Neural Network)

## Basic Algorithms

Searching Algorithms

- Linear Search
- Binary Search
- Jump Search
- Exponential Search

Sorting Algorithms

- Bubble Sort
- Selection Sort
- Insertion Sort
- Merge Sort
- Quick Sort
- Heap Sort

Graph Algorithms

- Dijkstra's Algorithm
- Bellman-Ford Algorithm
- Kruskal's Algorithm
- Prim's Algorithm –
- DFS (Depth-First Search)
- BFS (Breadth-First Search)

# PROBLEM

**Create an algorithm for:**

- A program adds two numbers.
- A program checks if a number is positive or negative.
- A program determines if a number is odd or even.
- A program calculates grade of student based on marks.
- A program to make a cup of tea.

## PSEUDO CODE

A pseudo code is  an informal way to describe a program. It can use natural language or compact mathematical notaion. Pseudo code is a rough sketch of the actual program.

**Pseudo code vs Algorithm**

- Algorithm is a step-by-step logical procedure to solve a problem whereas, pseudo code is a plain-language representation of an algorithm.
- Pseudocode begin with a START and ends with END whereas, an algorithm goes in between.

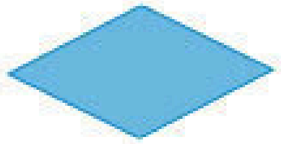
## EXAMPLE OF PSEUDO CODE

```
Start
Input three numbers: num1, num2, num3
If num1 > num2 and num1 > num3:
    Output "num1 is the largest"
Else if num2 > num1 and num2 > num3:
    Output "num2 is the largest"
Else:
    Output "num3 is the largest"
End
```

## FLOWCHART

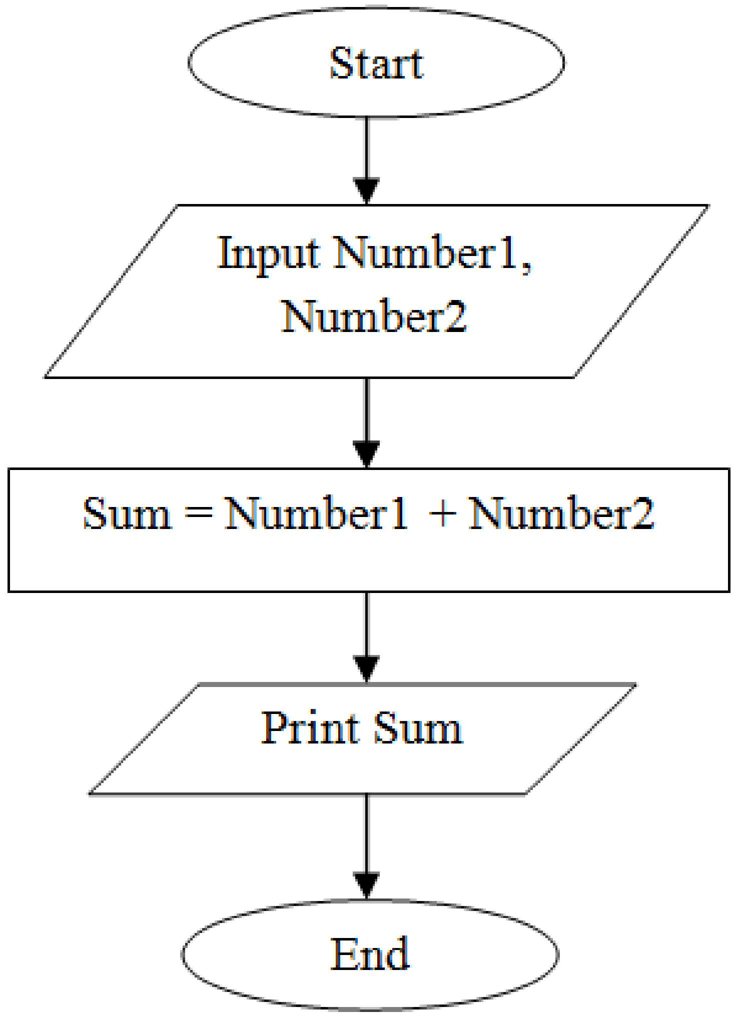
- Flowchart is a pictorial way to express algorithm or process.
- It uses symbols and arrows to depict the sequence of steps, decisions, and actions required to solve a problem or perform a task.
- Flowcharts are widely used in programming, business processes, and decision-making to simplify complex ideas.

# FLOWCHART - BASIC SYMBOL

| Symbol | Name | Function |
|---|---|---|
|  | Start/end | An oval represents a start or end point |
|  | Arrows | A line is a connector that shows relationships between the representative shapes |
|  | Input/Output | A parallelogram represents input or output |
|  | Process | A rectangle represents a process |
|  | Decision | A diamond indicates a decision |

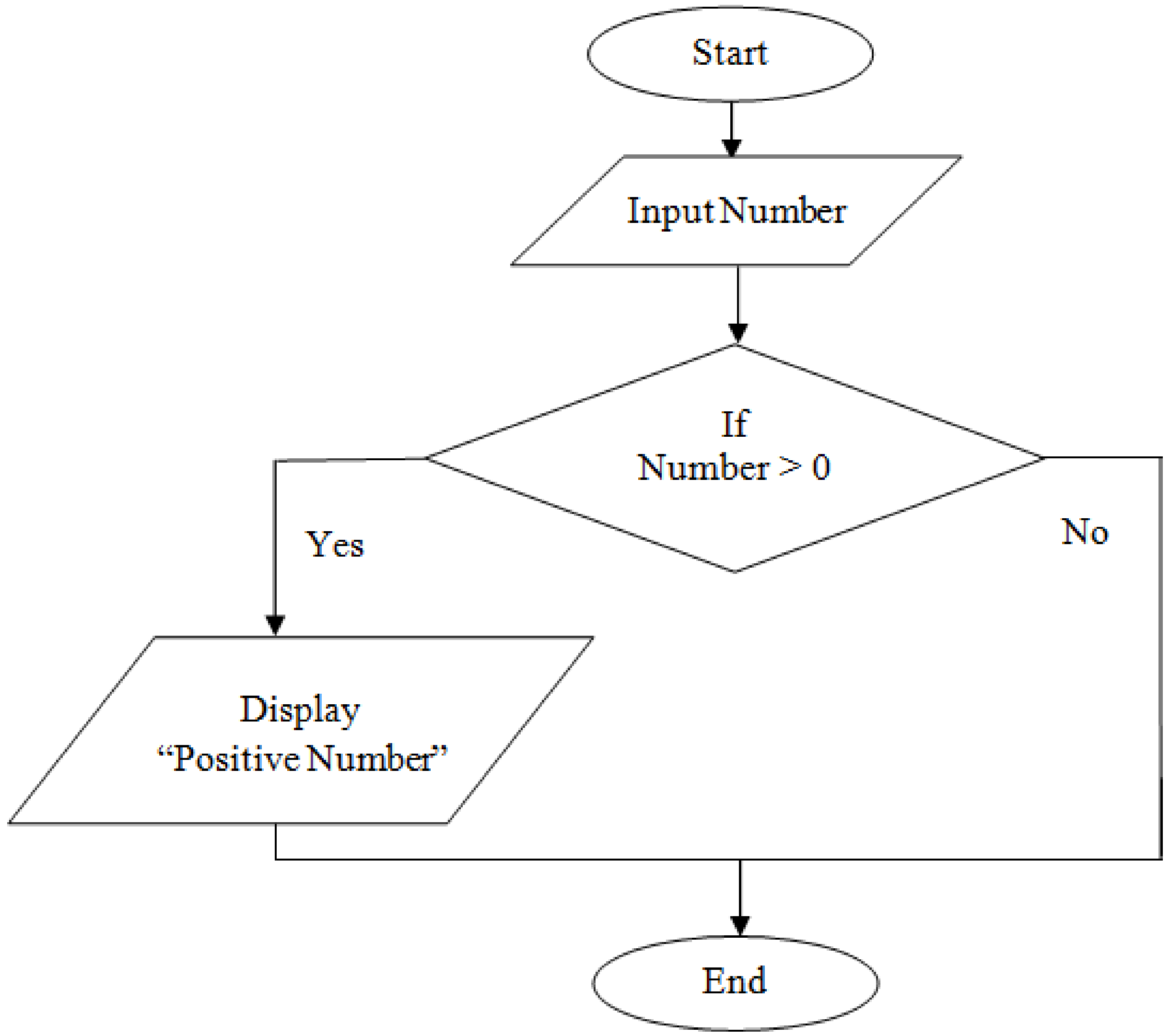
# EXAMPLE OF FLOWCHART

Problem: Add two numbers.

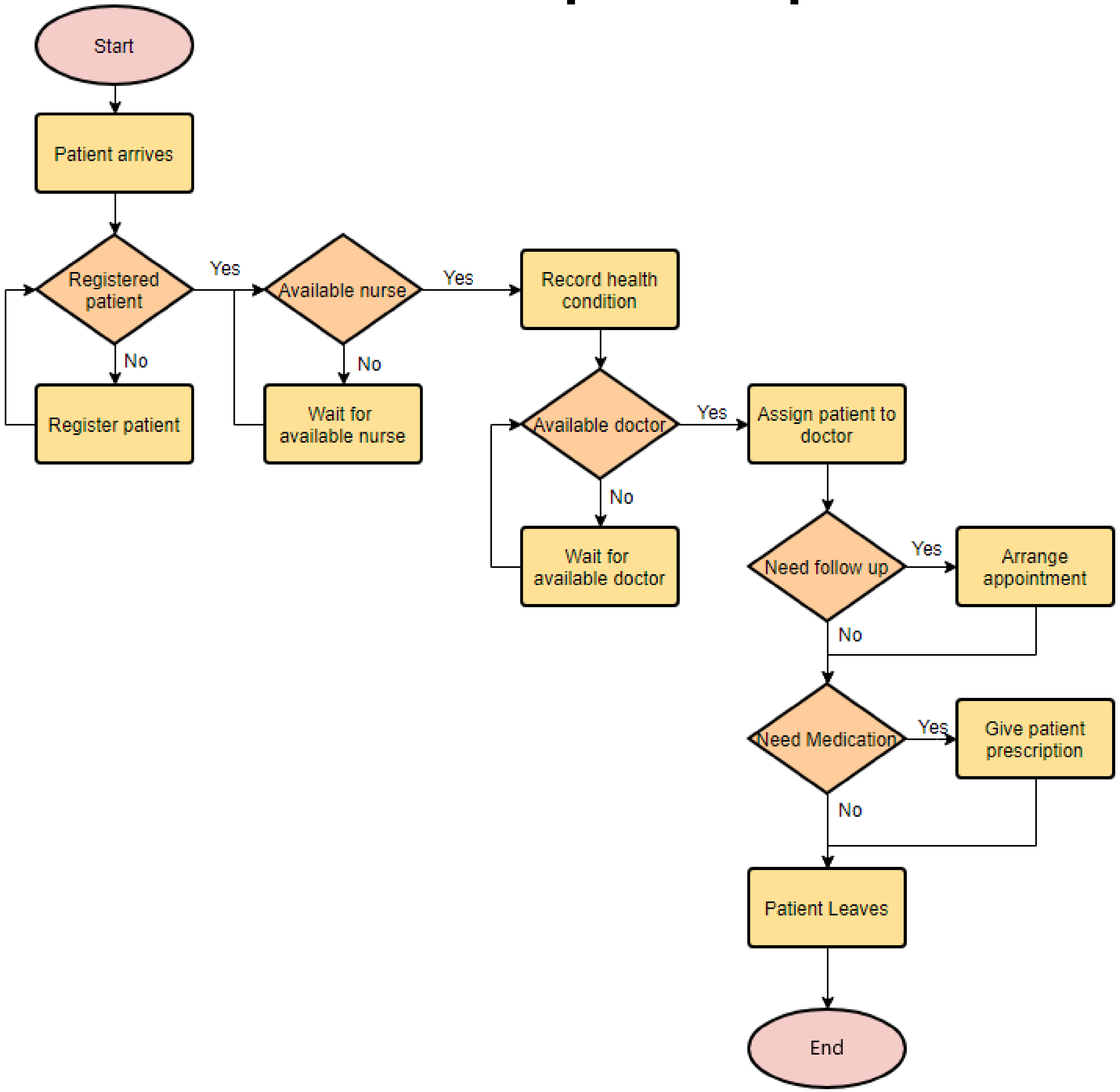


# EXAMPLE OF FLOWCHART

Problem: Check a number is positive or not.



# Flowchat Example: Hospital's Medical Services




Source: zenflowchart

## PROBLEMS OF FLOWCHART

Create flowchart for:

- A program adds two numbers.
- A program checks if a number is positive or negative.
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## PROBLEMS OF FLOWCHART

Create a flowchart that calculates the total price of a shopping bill. The program should ask user to input price and quantity of an item. Calculate the total price and check if the total exceeds Rs. 5000/-. If yes, apply 10% discount. If no, proceed without a discount. Display the final price.

## PROBLEMS OF FLOWCHART

Create a flowchart that simulates an ATM withdrawal. The program should input the balance and withdrawal amount. Check if the withdrawal amount is less than or equal to the balance. If yes, deduct the amount and print the remaining balance. If no, print an error message (insufficient funds).

## PROBLEMS OF FLOWCHART

A cinema offers discounts based on the viewer's age. Design a flowchart for a program that ask the user for their age and the ticket price. Apply a discount based on age. If the age is 60 or above, apply a 20% discount. If the age is between 5 and 18, apply a 10% discount. If the age is below 5, the ticket is free. Otherwise, no discount is applied. Display the final ticket price.

## PROBLEMS OF FLOWCHART

Create a flowchart for a Bike Rental System with Login Functionality. The system should include the following steps:

I. Prompt the user to log in by entering a username and password.
   a. If login is successful, proceed to the next step.
   b. If login fails, display error message and prompt user to try again.

II. Once logged in, ask user to
   a. Enter the rental period (in hours).
   b. Enter the user budget.

III. Calculate the total cost of renting the bike.
   a. Total Cost = 100 x Rental Period

IV. Compare the total cost with user budget.
   a. If budget is sufficient, display a confirmation message "Bike Rental Successful".
   b. If budget is insufficient, display an error message "Insufficient budget, try a shorter period".

# PROBLEMS OF FLOWCHART

Design a flowchart for a Movie Ticket Booking App. The system should:

1. Allow users to Register (if new) or Login (if they have an account).
2. After login, users can select a movie from the list and choose a time slot.
3. The system should ask the user for the number of tickets they want to buy.
4. Calculate the total ticket price based on the ticket price per person (e.g., Rs. 250/- per seat).
5. If the total exceeds Rs. 1000, offer a 10% discount. Otherwise, proceed without a discount.
6. The user proceeds to make a payment.
7. After successful payment, the system displays the final ticket details and download option for the e-ticket.

# THANK YOU